

Virtual Christmas Tree Implementation Checklist

Delivery window: September 8-23, 2026
Target: Production-ready launch before the October 1 first-wish unlock
Status: Planned

Purpose

Build an authenticated Control Hub experience where each team member places one personalized ornament on a shared Christmas tree, records one wish in each October-December release window, and lets teammates inspect unlocked wishes during the year-end call.

Industry Standard - Mental model: the browser renders the tree and collects intent; the server is the authority for identity, one-ornament ownership, safe placement, and whether a wish slot is unlocked. The database constraints and RLS policies remain the final enforcement layer.

Before deciding any **YOUR CALL**, predict the trade-off first. For each **HEADS-UP**, ask how that issue could have been detected earlier. Record accepted decisions in `docs/decisions/` with the decision, options, reasoning, and what would change the decision.

Ground Truth (September 8)

- **VERIFIED:** `pnpm typecheck` passed for all eight workspace projects on September 8, 2026.
- **VERIFIED:** `apps/web` contains authenticated self-service feature patterns, API routes, TanStack Query hooks, and Supabase Realtime subscriptions that this feature can reuse.
- **UNVERIFIED:** No dedicated Christmas-tree implementation was confirmed. The local `rg` executable is unavailable, so the planned discovery task must identify any overlapping event feature before implementation.
- **UNVERIFIED:** Supabase local migration and RLS integration coverage was not run for this planning pass.

Scope and Traceability

Requirement	Delivery coverage
US-01 / FR-01 / FR-02 / FR-03	Identity-backed ornament selection, one placement per user, responsive tree rendering, collision-safe coordinates, and visible ownership.
US-02 / FR-04	Three server-time-gated wish slots that retain access to earlier unlocked months.
US-03	Ornament-inspection modal with submitted wishes and locked-slot teasers.
FR-05	Read-only fullscreen presentation mode for the Christmas call.
Optional realtime	Realtime subscriptions refresh the shared tree when ornaments or wishes change.

Timeline

Date	Milestone	Exit evidence
Sep 8	Scope, decisions, and technical spike	Confirm route, auth source, seasonal event configuration, asset licensing, and Realtime eligibility.
Sep 9-10	Schema and access controls	Migration applies locally; constraints and RLS integration tests pass.
Sep 11-12	Server contract	Authenticated ornament and wish APIs reject invalid, duplicate, and locked operations.
Sep 15-17	Core team experience	A member can select, place, revisit, and inspect ornaments on desktop and mobile.
Sep 18-19	Presentation and realtime	Read-only fullscreen view and live updates work without exposing private drafts.
Sep 22	Quality and acceptance	Unit, API, E2E, accessibility, and responsive checks pass.
Sep 23	Launch readiness	Production configuration, monitoring, rollback procedure, and stakeholder sign-off are complete.

Phase 0 - Decisions and Feature Boundary (Sep 8)

- ☐ **0.1** Identify the owner route under `apps/web/src/app/(employee)` and reuse the closest authenticated interactive feature's API, hook, and component patterns. **Effort:** S. **Requirement:** FR-01. **Skip risk:** Feature is built outside portal conventions and incurs integration rework. **MEDIUM.**
- ☐ **0.2** Define a season/event configuration with `event_key`, `timezone`, `opens_at`, `wish_1_opens_at`, `wish_2_opens_at`, `wish_3_opens_at`, and `closes_at`; use `Asia/Manila` unless the event owner approves another timezone. **Effort:** S. **Requirement:** FR-04. **Skip risk:** Ambiguous date boundaries unlock wishes at inconsistent times. **HIGH.**
- ☐ **0.3 YOUR CALL** Use existing authenticated Control Hub accounts as the sole identity source. Do not ship a client-side name-claim alternative: it cannot reliably enforce one ornament per person. Only add an invitation/name-claim flow if external participants are a confirmed requirement, backed by verified email or a signed invitation. **Effort:** S. **Requirement:** FR-01. **Skip risk:** Duplicate or impersonated ornaments undermine the core promise. **HIGH.**
- ☐ **0.4 YOUR CALL** Decide who can see wishes before the Christmas call: recommended default is all authenticated teammates can inspect only submitted, already-unlocked wishes; admins can moderate entries; no role sees future-slot content. Record any stricter visibility policy in an ADR. **Effort:** S. **Requirement:** US-03. **Skip risk:** Privacy expectations are implemented inconsistently. **HIGH.**
- ☐ **0.5** Select and document 4-6 licensed or original ornament assets with alt text, stable asset keys, and dark/light contrast checks. **Effort:** S. **Requirement:** FR-02. **Skip risk:** Launch is blocked by unclear asset rights or inaccessible choices. **MEDIUM.**

Phase 1 - Data Model and Security (Sep 9-10)

- ☐ **1.1** Add a timestamped migration for `christmas_tree_events`, `christmas_ornaments`, and `christmas_wishes`, including standard audit columns and foreign keys to authenticated users. Keep `asset_type` constrained to the approved set. **Effort:** M. **Requirement:** US-01, US-02, FR-02, FR-04. **Skip risk:** Invalid assets, orphaned data, and seasons that cannot be configured safely. **HIGH.**

- ☐ **1.2** Enforce one ornament per user per event with a unique database constraint; enforce one wish per ornament and slot with a unique constraint and restrict `slot_number` to 1-3. **Effort:** S. **Requirement:** US-01, US-02. **Skip risk:** Concurrent requests create duplicate ornaments or wishes. **HIGH.**
- ☐ **1.3** Store positions as normalized percentages (`position_x`, `position_y`) with database bounds and reserve placement generation to the server. **Effort:** M. **Requirement:** FR-03. **Skip risk:** Ornaments render off-tree, overlap severely, or drift across viewports. **HIGH.**
- ☐ **1.4** Add RLS policies: authenticated users can read ornaments for the active event, create/update only their own ornament and wishes, and cannot change `user_id`, event, slot, or submission timestamps. Limit event configuration and moderation to approved admin roles. **Effort:** L. **Requirement:** FR-01, FR-04. **Skip risk:** A user can alter another teammate's ornament or bypass ownership. **CRITICAL.**
- ☐ **1.5 HEADS-UP** Enforce the slot release time in a database RPC or server-side transaction using database/server time, never the browser clock. The UI lock is presentation only; direct HTTP requests must receive `403` before the configured release time. **Effort:** M. **Requirement:** US-02, FR-04. **Skip risk:** Users bypass time gates by changing local time or replaying requests. **HIGH.**
- ☐ **1.6** Add Supabase migration/RLS integration tests for constraints, cross-user access denial, malformed positions, and each slot boundary. **Effort:** M. **Requirement:** US-01, US-02, FR-01, FR-03, FR-04. **Skip risk:** Database-layer guarantees remain untested. **HIGH.**

Phase 2 - Server Contract and Observability (Sep 11-12)

- ☐ **2.1** Create Zod-validated API routes or server actions for event snapshot, ornament placement/update, and wish upsert. Authenticate every request with the existing server Supabase client. **Effort:** L. **Requirement:** US-01, US-02, US-03. **Skip risk:** Invalid payloads or unauthenticated requests reach the data layer. **HIGH.**
- ☐ **2.2** Implement deterministic collision-aware position allocation: select from server-owned tree-safe zones, reject collisions at the ornament's responsive radius, and return a retryable conflict only when capacity is exhausted. **Effort:** M. **Requirement:** FR-03. **Skip risk:** The shared canvas becomes unreadable as participation grows. **HIGH.**
- ☐ **2.3** Return a single typed snapshot containing the active event, the requesting member's ornament/wish edit state, public ornament summaries, and server-computed slot availability/teaser labels. Do not return future wish content. **Effort:** M. **Requirement:** US-02, US-03. **Skip risk:** UI logic duplicates authorization rules or leaks locked wish text. **HIGH.**
- ☐ **2.4** Add audit-log events for placement, asset changes, wish submission/edit, admin moderation, and event configuration changes. Exclude wish content from logs unless retention and access requirements explicitly permit it. **Effort:** M. **Requirement:** FR-01, FR-04. **Skip risk:** Sensitive actions cannot be investigated without unnecessarily storing private wish text. **MEDIUM.**
- ☐ **2.5** Add structured error telemetry and basic adoption metrics: unique ornaments placed, slot completion by month, placement conflicts, rejected locked-slot requests, and API failures. **Effort:** S. **Requirement:** —. **Skip risk:** No baseline to diagnose launch issues or measure participation. **MEDIUM.**
- ☐ **2.6** Add API tests for authentication, one-ornament contention, ownership, validation, time gates in September/October/November/December, and public snapshot redaction. **Effort:** M. **Requirement:** US-01, US-02, US-03. **Skip risk:** Core contracts regress silently. **HIGH.**

Phase 3 - Team Experience (Sep 15-17)

- ☐ **3.1** Build the authenticated Christmas Tree page using existing `@hr-portal/ui` primitives, TanStack Query query-key conventions, loading states, error states, and optimistic mutations with rollback. **Effort:** L. **Requirement:** US-01, US-02. **Skip risk:** Slow or failed submissions leave the member without a reliable recovery path. **HIGH.**
- ☐ **3.2** Build the ornament selector with 4-6 assets, keyboard navigation, visible selection, name sourced from the authenticated profile, and a clear empty/missing-profile state. **Effort:** M. **Requirement:** US-01, FR-02. **Skip risk:** Users cannot reliably create a personalized ornament. **MEDIUM.**
- ☐ **3.3** Render the responsive tree using percentage coordinates and a stable ornament hit target; show owner identity on hover and keyboard focus without obscuring neighboring ornaments. **Effort:** L. **Requirement:** US-01, FR-03. **Skip risk:** Names are inaccessible or ornaments overlap on common screen sizes. **HIGH.**
- ☐ **3.4** Build three wish controls from the server-computed slot state: editable unlocked slots, retained prior slots in later months, submitted state, validation feedback, and visually disabled future slots with their exact release label. **Effort:** M. **Requirement:** US-02. **Skip risk:** The progressive-advent interaction is unclear or incorrect. **HIGH.**
- ☐ **3.5** Build the ornament detail modal with owner name, ornament asset, submitted eligible wishes, and locked-slot teaser states. Support pointer, keyboard, focus restoration, and screen readers. **Effort:** M. **Requirement:** US-03. **Skip risk:** The call-time reveal is inaccessible or exposes content prematurely. **HIGH.**
- ☐ **3.6** Add component tests for selector validation, locked/unlocked rendering, wish edit states, collision display, and detail-modal focus behavior. **Effort:** M. **Requirement:** US-01, US-02, US-03. **Skip risk:** UI regressions reach the event. **MEDIUM.**

Phase 4 - Presentation and Realtime (Sep 18-19)

- ☐ **4.1** Add a read-only fullscreen presentation route or mode with minimal chrome, dense readable ornament labels, responsive scaling, exit controls, and no mutation affordances. **Effort:** M. **Requirement:** FR-05. **Skip risk:** The Christmas-call host must use an unsuitable participant interface. **MEDIUM.**
- ☐ **4.2** Subscribe to allowed ornament and wish changes with the established Supabase Realtime hook pattern; invalidate/refetch the event snapshot and clean up channels on unmount. **Effort:** M. **Requirement:** Optional realtime. **Skip risk:** Shared-screen updates require refreshes and feel stale. **MEDIUM.**
- ☐ **4.3 HEADS-UP** Confirm Realtime replication and RLS behavior in staging before enabling it in production. A subscription must never receive another user's locked or unauthorized wish payload, even briefly. **Effort:** M. **Requirement:** US-02, US-03. **Skip risk:** Realtime becomes a data-disclosure path. **CRITICAL.**
- ☐ **4.4** Provide a polling/refetch fallback and a visible non-intrusive connection state for presentation mode. **Effort:** S. **Requirement:** FR-05. **Skip risk:** The tree appears frozen during a live call when Realtime is unavailable. **MEDIUM.**

Phase 5 - Acceptance, Release, and Handoff (Sep 22-23)

- ☐ **5.1** Run acceptance tests for every US/FR, including first placement, rejected duplicate placement, late-entry access to prior slots, future-slot denial, ornament inspection, and fullscreen mode. **Effort:** M. **Requirement:** US-01, US-02, US-03, FR-01 to FR-05. **Skip risk:** A stated requirement has no executable proof. **HIGH.**

- ☐ **5.2** Add Playwright coverage for employee placement and wishes, a second user inspecting the ornament, and admin presentation mode; cover desktop and mobile viewports. **Effort:** L. **Requirement:** US-01, US-02, US-03, FR-05. **Skip risk:** Critical cross-user flows fail after deployment. **HIGH**.
- ☐ **5.3** Perform accessibility checks for keyboard-only placement/inspection, modal focus trap/restoration, labels, reduced motion, contrast, and 200% zoom. **Effort:** M. **Requirement:** US-01, US-03, FR-05. **Skip risk:** Team members cannot participate or inspect wishes independently. **HIGH**.
- ☐ **5.4** Execute `pnpm lint`, `pnpm typecheck`, targeted Vitest suites, and relevant Playwright suites; record actual results in this document. **Effort:** M. **Requirement:** —. **Skip risk:** Type, formatting, and behavioral regressions ship. **HIGH**.
- ☐ **5.5** Configure production event dates, approved assets, enabled Realtime tables, feature navigation, observability alerts, and a rollback switch that disables the route without deleting data. **Effort:** M. **Requirement:** FR-02, FR-04, FR-05. **Skip risk:** The feature is technically complete but cannot be safely operated. **HIGH**.
- ☐ **5.6** Complete stakeholder walkthrough and publish a short host runbook covering presentation mode, moderation/escalation, support contact, timezone, and rollback. **Effort:** S. **Requirement:** FR-05. **Skip risk:** The event host cannot resolve issues during the Christmas call. **MEDIUM**.

Sequencing Constraints

Order	Work	Why it is sequenced here
1	Identity, event dates, schema, RLS	These define ownership and prevent time-gate bypasses before UI work starts.
2	Server contract, placement algorithm, telemetry	The client needs a trusted snapshot and safe coordinates; instrumentation needs a pre-launch baseline.
3	Participant UI	It can then consume one stable, protected contract.
4	Realtime and presentation mode	These extend the proven read model and should not block the core participation flow.
5	E2E, accessibility, release readiness	Cross-user behavior and call-time operation must be proven before October 1.

Hard constraint: Do not expose the Christmas Tree route to real users until Phase 1 RLS/time-gate tests and Phase 2 API authorization tests pass. Client-side disabled controls are not an authorization boundary.

Definition of Done

- ☐ Every requirement in the traceability table has a passing automated or documented acceptance check.
- ☐ One authenticated account can create and own exactly one ornament per event; concurrent duplicate creation is rejected.
- ☐ Only server-authorized slots can be created or edited, and future wish content is never returned to unauthorized clients.
- ☐ The tree remains legible and interactive on supported desktop and mobile viewports.
- ☐ Presentation mode works in fullscreen and receives safe live updates or its fallback refresh path.
- ☐ Production dates, assets, monitoring, rollback, and host runbook are approved by September 23.